

Visual Feature Attribution in UMAP Clustering Using The Explainable Boosting Machine

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Abstract

Dimensionality reduction techniques such as Uniform Manifold Approximation and Projection (UMAP) often yield visually discernible clusters, but the interpretation of these clusters can be difficult. This report presents a hybrid visualization tool that combines UMAP with the Explainable Boosting Machine (EBM) to identify which features of a dataset contribute to clustering in UMAP space. The system enables the extraction of interpretable insights from clustered 2D embeddings by training EBMs to predict cluster membership. The implementation follows the approach proposed in Salmanian et al. 2024 and is validated on datasets including synthetic 3D separable data, coronary artery disease (CAD) patient records, penguin physiology, and cellular breast cancer data in the context of classification. The resulting system provides an interpretable and effective visualization method for attributing features to visual clustering behavior.

1 Introduction

Dimensionality reduction (DR) techniques such as Uniform Manifold Approximation and Projection (UMAP) enable effective visualization of high-dimensional datasets by projecting them into low-dimensional space while preserving local and global structures. The main motivation for using DR tools like UMAP lies in the identification of (sometimes non-binary) classification-based clusters within complex data. Although UMAP excels at producing such clusters, it provides limited insight regarding the influence of individual input features on the resulting structure. This is a common downfall of DR techniques and is a highly researched field.

To address this interpretability gap, the Explainable Boosting Machine (EBM), a generalized additive model with pairwise interactions, is applied in conjunction with UMAP. The objective is to identify which input features influence UMAP-based clustering, thereby enhancing both the utility and trustworthiness of the visualization process.

The system implemented in this project is inspired by the methodology described in *DimVis: Interpreting Visual Clusters in Dimensionality Reduction With Explainable Boosting Machine*, which uses EBMs to infer and visualize feature importance with respect to 2D UMAP cluster locations. The scope of the implementation includes automation of the analytical workflow, generalization to various datasets, and the generation of visual outputs linking features to UMAP embeddings.

2 Related Work

The methodology for the UMAP-EBM tool in this project is based on the work of Salmanian et al. [1], which first proposed the use of EBMs to interpret clusters generated by UMAP. This

approach is based in the broader field of interpretable machine learning, where post hoc analysis techniques are used to explain the behavior of complex models like UMAP. Salmanian et al. [1] dives deeper into the outputs of the EBM, but the quality of insights found in this paper match those found by Salmanian’s team.

UMAP is known for its nonlinear embedding capability, but this same nonlinearity complicates the interpretation of clustering structures. Previous efforts have applied surrogate models, such as decision trees, to explain black-box models. However, EBMs provide a superior trade-off between interpretability and performance due to their additive structure and ability to model low-order interactions. This makes the EBM well-suited for identifying feature contributions in DR-based visualizations.

2.1 Data and Task Abstraction

An emphasis of this tool was dataset adaptability. UMAP-EBM can conduct a feature abstraction on any provided dataset that specifies a classification field. With this being said, a few example datasets are included within the tool to provide the user with guidance. These datasets include patient Coronary Artery Disease (CAD) data, physiological penguin data separated by species, and patient breast cancer data.

The datasets used in this system are structured as multivariate tables, where each row corresponds to an individual data item—a patient or a penguin—and each column represents an attribute, which may be either quantitative (e.g., age or measurement length) or categorical (e.g., symptom category or species label). In the UMAP processing, all data besides the classification field must be numeric (float / int) leaving features of quantitative ratio and interval with one nominal flag. These numeric datasets are initially expressed in matrix form, $X \in R^{n \times d}$, where n is the number of items and d the number of features – excluding the classification field.

According to the motivation of this project, it is most reasonable to claim discover: correlation as the primary task abstraction. The point of the EBM in simplest terms is to identify which data features cause clusters in DR space. Clusters can be thought of as ”effects” from the UMAP DR that the EBM attempts to identify ”causes” for. As such, UMAP-EBM discovers correlation between data features and a specific graphical clustering.

A more extensive description of the vast amounts of data included in this project is a waste of time.

3 Solution

The solution to the problem of interpretability loss from dimension reduction techniques is to simply run DR clusters through the EBM. UMAP is a non-linear dimensionality reduction technique that constructs a weighted graph from high-dimensional data and optimizes a low-dimensional layout that preserves both local and global structure. This enables the projection of a high-dimensional dataset into a two-dimensional space where similar items are positioned close to one another, often forming visually interpretable clusters. However, UMAP does not provide insight into which input features influence the positioning of points in this reduced space. To address this, EBM is used. EBM is a type of generalized additive model that learns transparent, low-order interactions between features and a target variable, making it well-suited for post hoc explanations of machine learning outputs.

The full system operates as follows. First, the dataset is loaded and preprocessed into the appropriate format: a numeric data matrix $X \in R^{n \times d}$, stripped of non-numeric features, and an associated label vector $Y \in R^n$, typically used for coloring or validating classifications. UMAP

is then applied to X , producing a 2D embedding that is visualized on a cartesian grid. Points are colored according to their class labels, often revealing classification based spatial clusters. A user can then select a specific cluster from this visualization with the goal of understanding which features in the original high-dimensional space are responsible for the spatial separation of that group. To analyze this, a binary classification task is constructed: the selected points are labeled as class 1 (in-cluster), while the remaining points are labeled as class 0 (out-of-cluster). An EBM is trained on this binary classification task using the original features in X as predictors. Because of EBM's additive structure and interpretable modeling of feature effects, the output consists of ranked feature and feature pair importance scores. These are visualized directly in the notebook environment, allowing users to draw conclusions about which variables are driving the observed cluster structure in UMAP space.

4 Implementation

The entire tool is run in one Jupyter notebook: `auto_UMAP_EBM.ipynb`. The notebook is split into three distinct sections.

1. Dataset Selection and Processing
2. UMAP Dimension Reduction and Bokeh Plotting
3. EBM Interpretability Extraction

A description of each section as well as an explanation for why sections were separated accordingly is provided below.

4.1 Dataset Selection and Processing

This section is quite self explanatory. Whichever dataset the user desires to interpret is selected by saving/updating three variables in the notebook session: (1) the (local) path to the dataset, (2) the (sometimes multi-state) classification column of interest labeled `flag`, and (3) The name of the column in the dataset that functions as a row id. If no such column exists, the user must set `rowid = None`.

The author mandates the inclusion of a `flag` argument to motivate cluster selections. A row id must be identified so it can be removed as described in the processing phase. Internal datasets – like patient breast cancer cases – are handled differently because they are often stored in dictionaries instead of the more widely supportable csv format that allows for maximum flexibility.

After identifying the three variables, data (besides internally stored python data) is loaded using the pandas csv reader. The dataframe is stripped of N/A values, non-numeric fields, and any additional row id columns. This numeric data matrix is mean centered and standardized to comply with UMAP data requirements. Centering at the origin and standardizing feature scales prevent features with larger values and non-centered means from dominating distance calculations. Finally, a dataframe is displayed for the user's benefit.

This section stops here to prevent redundant computing, make it easy for the user to switch between datasets, and present a preview of the data matrix before dimension reduction.

4.2 UMAP Dimension Reduction and Bokeh Plotting

The UMAP algorithm is implemented in 2 concise lines. The first line, `reducer = umap.UMAP()`, allows the user to customize the reducer by changing UMAP parameters. This was not an emphasis of the project, so the default reducer was used. The second line, `embedding = reducer.fit_transform(scaled_df_numeric)` transforms the scaled data matrix into $R^{n \times 2}$ UMAP space.

The remainder of this section details the interactive 2D UMAP plot supported by the python visualization package Bokeh. After a data source is specified and a color coding system is computed, the figure is plotted including a lasso selection tool and `save_indices` feature. Together, these Bokeh features allow the user to select a cluster to be run through the EBM.

4.3 EBM Interpretability Extraction

Once a user confirms the selection of a cluster by clicking the white "Save Selection" button below the Bokeh plot, the EBM classification algorithm can be run. Running the EBM code with an empty `save_indices` vector can have severe consequences and may even ruin the entire python session. This has occurred when the user notices the EBM code only displays the accuracy score and classification report when the cell has completed. Solutions to this problem involve restarting the kernel, deleting the `auto_UMAP_EBM` tab from the IDE, copying all code into a new notebook, and combinations of these suggestions.

The EBM code starts by assigning a '1' and '0' to data within and outside the selected cluster respectively. In proper ML convention, the algorithm is trained and tested on an 80% / 20% train / test split. The EBM works by iteratively updating functions that map a feature or a pair of features to the observed classification (1 or 0). Feature functions are found independently from one another, so this algorithm is run in parallel. The nonlinearity and parallelization of this algorithm makes it extremely flexible and effective.

This section outputs statistics pertaining to the model accuracy. Two areas of attention are the precision and recall of the EBM. Precision measures the accuracy of predictions, while recall measures the model's ability to identify all cases in the respective category. These statistics are sensitive to the size of selected clusters. If a small percentage of points are selected or excluded from the cluster, correctly identifying them would be hard. The main output of the EBM is the horizontal bar plot measuring the "weighted mean absolute score" of relevant features and feature interactions that reflect just how influential they are in UMAP-ing points to the cluster.

5 Solution Test Case: 3D Demo

Before discussing the results to erroneous UMAP-EBM applications, it is wise to verify the tool's ability to correctly identify the relevant features causing classification clusters in UMAP space. Thus, a controlled test system was created. The basis of this test was to linearly separate random 3D data with a vertical plane only dependent on two spacial features x_1 and x_2 . A successful UMAP-EBM tool would identify that classification depends only on the two spacial features.

$N = 100$ random points were created in $(-1, 1)^3 \in R^3$ to be separated by a vertical plane defined by the normal vector $p = [0.2, -0.5, 0.5, 0]$. Notice that $p_4 = 0$ forces the decision boundary to be vertical. In other words, the decision plane does not depend on the third dimension x_3 . Classification labels are calculated by taking the dot product of each item x_i with the plane p as follows:

$$y_i = p \cdot x_i = [0.2, -0.5, 0.5, 0] \cdot [1, x_{i,1}, x_{i,2}, x_{i,3}]^T.$$

Notice that the zero in p_4 removes x_3 's influence on the classification. The selection of p makes the data classifiable by a vertical plane in R^3 and by a line when projected along x_3 into R^2 :

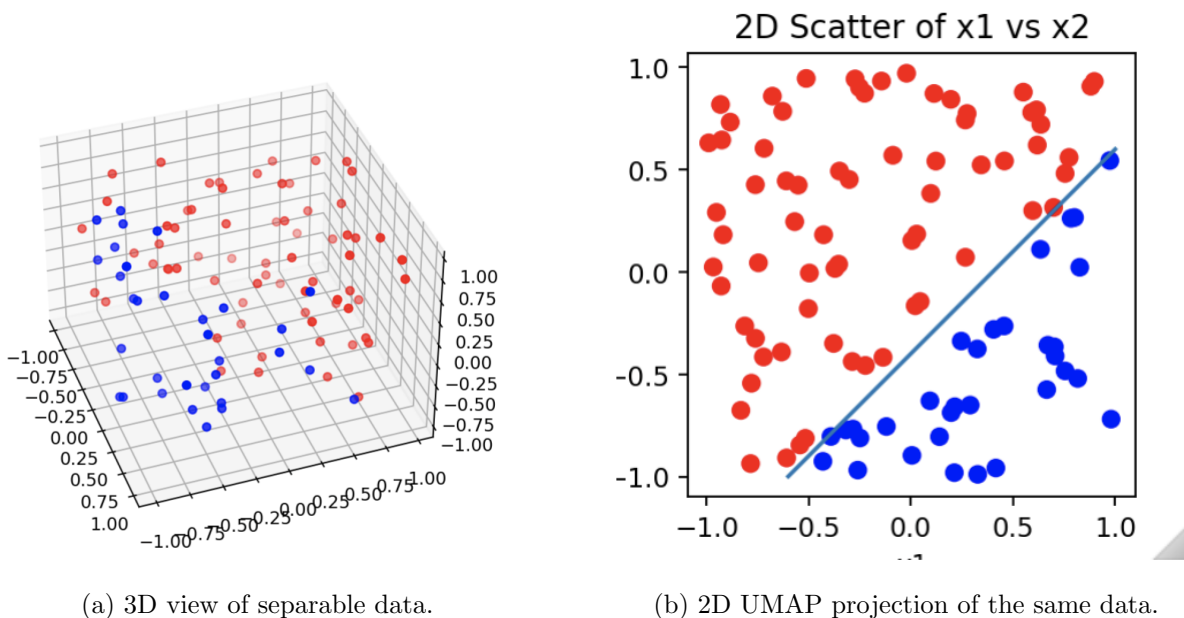


Figure 1: The 2D projection of a 3D dataset that is separable by a vertical plane. The projection remains linearly separable.

Classification independence of x_3 has been verified. The next step is to submit the 3D data and classifications into the UMAP-EBM tool without knowledge of the boundary p . If the tool functions correctly, it will suggest that classification depends on x_1 and x_2 far more than x_3 .

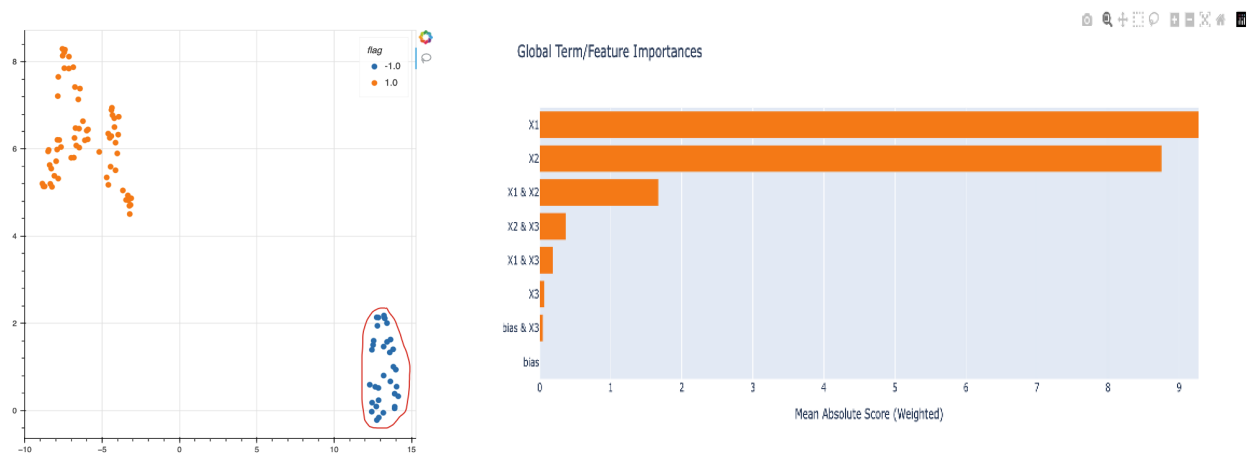


Figure 2: UMAP-EBM identifies that classification is independent of x_3 supporting the enforced condition of a vertical decision boundary plane.

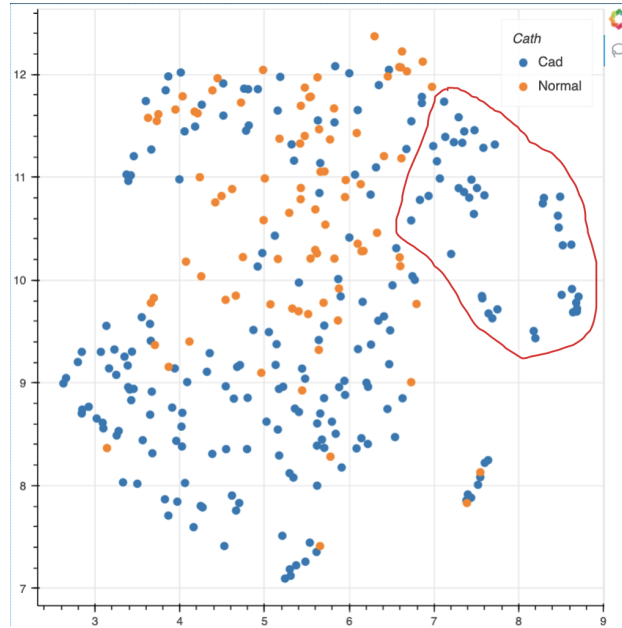
Immediately, UMAP creates distinct classification based clusters. Running the EBM on either of the clusters reveals the predicted outcome: the EBM has identified that the main features/feature

pairs determining classification are x_1 , x_2 , and x_1 & x_2 . Thus, the UMAP-EBM tool accurately identifies cluster defining features.

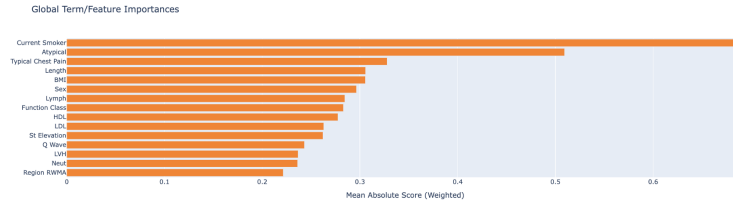
6 Results

UMAP-EBM was applied to the three datasets running the EBM on various UMAP clusters. Both UMAP and the EBM are non-deterministic models which prevents reproducibility. This can be seen as a positive, stimulating different UMAP projections that result in alternative feature abstractions. For each dataset, the selected UMAP region provided the respective EBM explanation.

6.1 Coronary Artery Disease



(a) A CAD cluster is selected on the right wing of the UMAP projection.



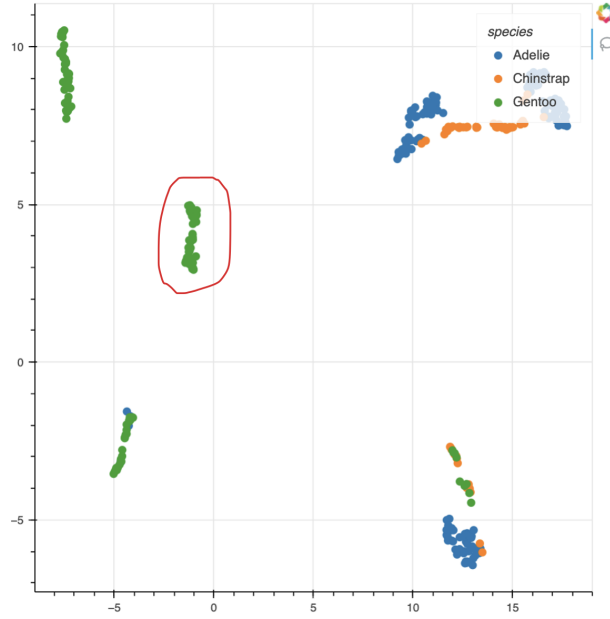
(b) The selected cluster reflects patients who currently smoke and experience typical/atypical chest pain.

Figure 3: Feature attribution for a visually selected cluster in the CAD UMAP projection. (a) Cluster selection in UMAP space. (b) EBM interpretation showing relevant features.

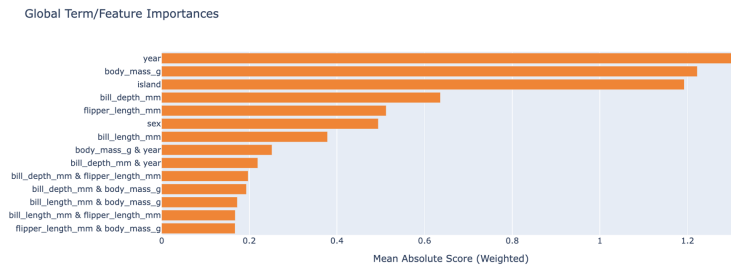
As illustrated in Figure 1a and 1b, the selected CAD cluster correlates to patients that currently smoke and experience both typical and atypical chest pain. Currently smoking is the main feature that maps to this area in UMAP space. This makes sense because smoking is a leading cause of heart disease like coronary artery disease. Since this cluster only encompasses CAD patients, it

makes sense to pick up one of the leading causes of CAD as an explainable feature. In previous iterations, the author identified clusters predicated mostly on a family history of CAD reflecting the fact that CAD is hereditary. This example immediately illustrates how useful UMAP-EBM is in revealing correlations within high-dimensional datasets.

6.2 Penguin Physiology



(a) A Gentoo Penguin cluster is selected on the left side of the UMAP projection.

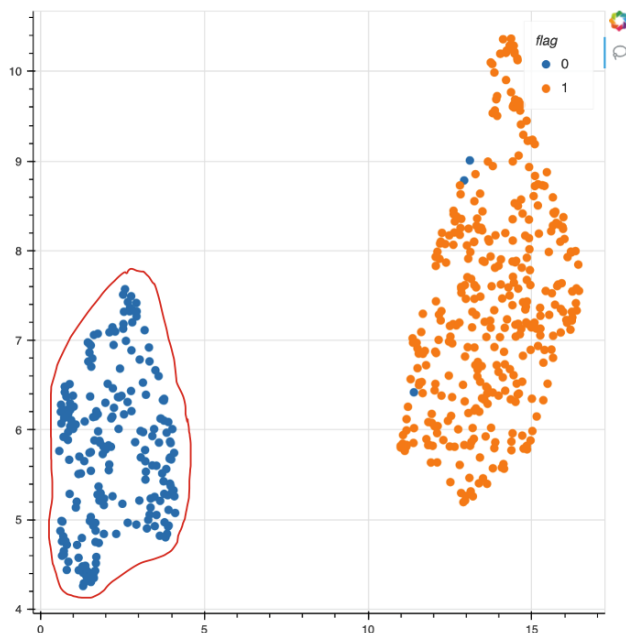


(b) The selected cluster reflects penguins who were surveyed in a specific year. The main characteristic defining Gentoo penguins appears to be body mass.

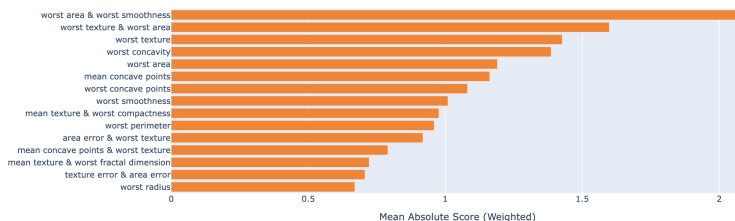
Figure 4: Feature attribution for a visually selected cluster in the Penguin UMAP projection. (a) Cluster selection in UMAP space. (b) EBM interpretation showing relevant features.

The seemingly ambiguous separation of Gentoo penguins in UMAP space can be attributed to varying survey years. Other defining qualities like body mass, island location, and bill depth are revealed as cluster features.

6.3 Breast Cancer



(a) The non-cancer cluster is selected on the left side of the UMAP projection.



(b) The main features that predict breast cancer are feature pairs including worst area & worst smoothness

Figure 5: Feature attribution for a visually selected cluster in the Breast Cancer UMAP projection. (a) Cluster selection in UMAP space. (b) EBM interpretation showing relevant features.

The UMAP projection creates two nearly perfect clusters. When considering the dataset, this makes sense. The features of sklearn’s `load_breast_data()` dataset measure statistical data pertaining to breast cells. Some features like worst area measure the average radii of the largest observed cross-sectional area of a cell’s nucleus. Since cancer is a cellular disease, patterns in this data are extremely consistent in cancer patients which is why such clear classification clusters arise in UMAP space.

7 Discussion

The visualization system provides a robust means for linking input features to clusters in reduced-dimensional space. The use of EBMs ensures that both global and local attributions are interpretable. This approach serves well in both exploratory analysis and domain-specific applications. User flexibility was emphasized to encourage experimentation.

However, certain limitations are inherent to the approach. Cluster generation depends on

unsupervised algorithms whose behavior is sensitive to parameter settings. Additionally, while EBMs offer transparency, their nonlinearity makes them prone to overfitting.

Future improvements could involve extending compatibility with more DR algorithms including supervised DR methods. Other considerations would be exploring more interpretability techniques like support vector machine and neural networks.

8 Conclusion

A hybrid visualization system has been implemented that integrates UMAP for dimensionality reduction and EBMs for interpretability. The tool supports various datasets and effectively highlights the relationship between input features and visual cluster formation. This methodology demonstrates a valuable approach for interpretable visual analytics, particularly in high-dimensional domains, and leads to the discovery of valuable insights.

References

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